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Lema de Riemann-Lebesgue en $\mathcal{L}^1(\mathbb{R})$

Lema 1 (de Riemann-Lebesgue). *Sea* $f \in \mathcal{L}^1(\mathbb{R})$*, entonces* $\lim_{|\xi|\to\infty} |f^\wedge(\xi)| = 0$.

***Demostración*:** Aplicamos la identidad de Euler $e^{-i\pi} = -1$:

$$f^\wedge(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x\xi}\,\mathrm{d}x = -\int_{\mathbb{R}} f(x) e^{-2\pi i x\xi} e^{-i\pi}\,\mathrm{d}x = -\int_{\mathbb{R}} f(x) e^{-2\pi i \left(x+\frac{1}{2\xi}\right)\xi}\,\mathrm{d}x.$$

Ahora hacemos el cambio de variable $y = x + \frac{1}{2\xi}$, con lo que

$$f^\wedge(\xi) = -\int_{\mathbb{R}} f\left(y - \frac{1}{2\xi}\right) e^{-2\pi i y\xi}\,\mathrm{d}y.$$

$$\implies 2f^\wedge(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x\xi}\,\mathrm{d}x - \int_{\mathbb{R}} f\left(y - \frac{1}{2\xi}\right) e^{-2\pi i y\xi}\,\mathrm{d}y$$
$$\implies f^\wedge(\xi) = \frac{1}{2}\left[\int_{\mathbb{R}} \left(f(x) - T_{\frac{1}{2\xi}} f(x)\right) e^{-2\pi i x\xi}\,\mathrm{d}x\right]$$

Por tanto, al calcular la norma, llegamos a que

$$|f^\wedge(\xi)| \leqslant \frac{1}{2}\int_{\mathbb{R}} \left|f(x) - T_{\frac{1}{2\xi}} f(x)\right|\mathrm{d}x = \frac{1}{2}\left\|f - T_{\frac{1}{2\xi}} f\right\|_1.$$

Por el `lem-convergencia-lp-traslacion/`**??**, concluimos que $|f^\wedge(\xi)| \xrightarrow[|\xi|\to\infty]{} 0$. ∎

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